

Bispecific antibody treatment

Bispecific antibody drugs are a type of cancer treatment that uses your white blood cells to find and target cancer cells. This page explains how bispecific antibody treatment works, what to expect and possible side effects.

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What are bispecific antibodies?

Bispecific antibodies are a type of cancer drug treatment. They are mainly used to treat blood cancers such as [myeloma](#), [lymphoma](#) and [leukaemia](#).

Bispecific means the drug can attach to 2 different cells at the same time:

- One part of the drug attaches to the cancer cell.
- The other part attaches to a type of white blood cell called a T-cell. These are cells in the blood that help fight infection and disease.

Image: A bispecific antibody links a cancer cell and a T-cell. One arm attaches to the cancer cell and the other arm attaches to the T-cell. Long description: The diagram shows how a bispecific antibody attaches to a cancer cell and a T-cell at the same time. On the left is a large circle representing a cancer cell. On the right is another large circle representing a T-cell. Between the 2 cells is a Y-shaped structure labelled "Bispecific antibody". The top 2 arms of the Y shape are different to each other. One has a square shape that connects to a square on the surface of the cancer cell. The other has a triangle shape that connects to a triangle on the surface of the T-cell.

The drug holds the cancer cell and the T-cell together. This helps the T-cell recognise and attack the cancer cell.

Other names for bispecific antibodies

This type of treatment may be described in different ways. Your cancer team might use the name or brand name of your drug. For example:

- [blinatumomab](#) (Blincyto®)
- [elranatamab](#) (Elrexfio®)

- [epcoritamab](#) (Tepkinly®)
- [glofitamab](#) (Columvi®)
- [teclistamab](#) (Tecvayli®).

Other terms to describe bispecific antibodies include:

- bispecific monoclonal antibodies
- BsAb or BsmAb
- bispecific T-cell engager or BiTE
- targeted therapy or immunotherapy.

Your course of bispecific antibody treatment

Your cancer team will discuss your bispecific antibody treatment plan and possible side effects with you before you agree ([consent](#)) to have treatment.

How often you have treatment, how it is given and how long your course of treatment lasts depends on:

- the type of cancer
- the bispecific antibody you are having
- how the cancer responds to the treatment
- any side effects you have.

Bispecific antibody drugs are given as several sessions of treatment. You have rest periods in between each session of treatment – this is when you have no treatment. The rest period allows your body to recover from the side effects. The treatment session and rest period make up one cycle of treatment. Your cancer team can explain how many cycles you may need.

How bispecific antibody drugs are given

Some bispecific antibody drugs are given as a drip into a vein. This is called an intravenous infusion. Others are given as an injection under the skin. This is called a subcutaneous injection.

Bispecific antibody drugs may be given with other cancer drugs. Your cancer team can explain what to expect.

You will usually stay in hospital for your first few treatments so your cancer team can monitor you closely. You may be able to have later treatments at a day unit or clinic as an outpatient.

Side effects of bispecific antibody treatment

Your cancer team will give you information about possible side effects before you start treatment. You may get some of the side effects but you are unlikely to get all of them.

Different drugs can cause different side effects. You can search for information about a specific drug or combination of drugs using our [cancer treatments and drugs A to Z](#).

Contacting the hospital

Your cancer team will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Sometimes side effects can become serious very quickly. Always contact the hospital straight away for advice.

Immune system side effects

Some side effects happen because bispecific antibodies make the immune system active. You can usually manage these safely with support from your cancer team. There are treatments that can help.

Side effects are important to know about because they can become serious without treatment. They include the following.

Cytokine Release Syndrome (CRS)

This side effect happens because the bispecific antibody drug makes the body release large amounts of cytokines. Cytokines are chemical messengers that help the body fight infection and illness.

You may have flu-like symptoms while you are having the drug, or hours or days later. Symptoms can range from mild to severe. You may need treatment and close monitoring in hospital until they improve.

CRS is most likely to happen the first few times you have a bispecific antibody. But it can also happen later. If you have any of the following symptoms during or after treatment, contact the hospital straight away on the 24-hour number:

- a temperature of 38°C or above
- feeling hot or flushed
- chills or shivering
- a skin rash
- a fast heart rate
- feeling dizzy or sick
- a headache
-

muscle or joint pain

- difficulty breathing or shortness of breath.

Effects on the brain and ICANS

Rarely, bispecific antibodies can cause temporary effects on the brain. This is called neurotoxicity or immune effector cell-associated neurotoxicity syndrome (ICANS).

Side effects are usually mild – for example, handwriting changes or feeling confused. But they can become serious without treatment and support. Some people will need treatment in intensive care until the symptoms improve.

ICANS is most likely to happen the first few times you have a bispecific antibody. But it can also happen later.

If you have any of the following symptoms after you go home, contact the hospital straight away on the 24-hour number:

- difficulty speaking, writing or doing fiddly tasks, such as fastening buttons or tying shoelaces
- drowsiness or extreme tiredness
- memory loss, confusion or disorientation
- tremors, shaking or muscle weakness
- loss of balance
- headaches
- seizures (fits).

You must not drive if you have these side effects. Your cancer team may also tell you not to drive or use heavy machinery when you start bispecific antibody treatment. Follow their advice carefully.

Risk of infection

You are more likely to get [infections](#) during and after finishing bispecific antibody treatment. You also have a higher risk of serious illness from infection.

To reduce your risk of getting an infection, your cancer team may give you:

- antibiotics and other drugs to take during your treatment
- ongoing antibody injections or drips.

It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

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feeling shivery and shaking

- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Tumour flare reaction

Bispecific antibody treatment can cause a sudden, temporary worsening of cancer-related symptoms at the start of treatment. This is called tumour flare reaction.

Always tell your cancer team if you notice any new or suddenly worsening symptoms. They can give you painkillers and other drugs to help.

Long term and late effects of bispecific antibodies

Cancer treatments sometimes cause side effects that do not go away after cancer treatment ends. Or they may cause side effects that start months or years after treatment ends. These are called long term or late effects.

Bispecific antibodies are newer treatments, and possible long term effects are still being researched. The benefit of having the treatment usually outweighs the risk of developing serious long term effects.

Your cancer doctor, specialist nurse or pharmacist can explain what is known about your treatment. Always get advice from your cancer team about any new symptoms or side effects.

Low antibody levels

The most common long term side effect of bispecific antibody treatment is low antibody levels. This is called hypogammaglobulinaemia. Antibodies are proteins released by white blood cells to help fight infection and illness.

Your cancer team will check your antibody levels during treatment. If you have low antibody levels:

- you are more likely to get infections
- vaccinations are likely to be less effective.

This side effect usually gets better with time. But some people may need treatments to help reduce their risk of infection, such as:

- antibiotics and other drugs to take during treatment
- regular antibody treatment given as a drip into a vein – this is called intravenous immunoglobulin treatment.

More information about bispecific antibodies

If you need information about a specific bispecific antibody, you can check our [cancer treatments and drugs A to Z](#). We have detailed information about many types of cancer drugs and their possible side effects.

You can also talk to your cancer team if you want more detailed information about a treatment. Or visit the [electronic medicines compendium \(emc\)](#), which has patient information leaflets (PILs) for individual drugs.

Other useful resources

If you want to explore more information about treatment and side effects, you can visit our [cancer treatment forum](#). Our Online Community is a place to talk with others having treatment, share your experience, and ask questions.

We also have information about treatment in our booklets:

- [Ask about your cancer treatment](#)
- [Side effects of cancer treatment](#)
- [Making treatment decisions.](#)

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our cancer drug treatment information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Chaganti S, Dulobdas V, Wilson MR, Tucker D, Townsend W, Parry-Jones N, et al. Clinical management of bispecific antibody therapy for lymphoma: A British Society for Haematology Good Practice Paper. Br J Haematol. 2025; 207(4): 1227–1241. Available from www.doi.org/10.1111/bjh.70018 [accessed August 2025].

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Get in touch

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



0300 1000 200

Our Support Line is available Monday to Friday, 9:30am-4:30pm



Email us

Email the Macmillan Fundraising team if you have any questions.



Online Community



Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.